CHAPTER 10

Putting It All Together

10.1 EXECUTIVE SUMMARY

In this chapter, we review the characteristics of the applications described in this book. The characteristics of the applications are manifested in the form of a database a query interface to which is available on this book's companion website. This chapter also includes some sample figures that depict the relationships between applications and their characteristics.

10.2 INTRODUCTION

The basic premise of this book has been that in order to build compelling new virtual reality applications it is wise to study the characteristics of extant applications. By studying other applications, the VR experience developer is able to draw on ideas that are suitable for their new application, while avoiding the pitfalls of weaknesses that other VR application developers have encountered.

Throughout this book, we have pointed out the various characteristics of dozens of VR applications that are foundational to the field. Applications are categorized by their genre, their hardware, their field of application (such as medicine, business, education, etc.), and various other categorizations. Additionally, we have noted the existence and nonexistence of various traits of each application. For example, we have noted whether an application uses audio or not, and if so, whether the audio is placed in three dimensions or not, etc. Another example is whether the application included haptic feedback, and if so, whether it is a tactile response or kinesthetic feedback.

All of these characteristics can be gleaned by reading the application descriptions included in this book. However, in order to make this book especially useful as a reference resource, we have encoded the characteristics of each application into an online database that supports queries by application developers to quickly allow them to drill into the relevant information from applications that provide the characteristics that they are interested in. So, for example, if an application developer is interested in including audio in their new application, they can query the database and find which applications described in the book use audio. The database supports much more complex queries as well, such as "Show me all of the applications that use 3D audio that also use head-mounted displays, but don't include haptic feedback." Conversely, it is also possible to query the database to learn the characteristics of the applications described in this book. For example, one could query the database to learn the characteristics of the *Officer of the Deck* application.

10.3 THE DATABASE FIELDS

In order to use any database to its maximum potential, it is important to know the types of data that it includes and what is meant by each data element. A full listing of the database fields is included in Chapter 2 and a definition of each field is available on this book's companion website.

The easiest way to explain the schema of the database is via an example or two. Continuing with the "audio" example, one can query for applications that use audio. Figure 10-1 shows the list of applications in this book that use audio.

Another level down, one can query for specific audio characteristics. In this database, subcategories of audio include "3D audio," "Nonspeech audio," "Sonification," "World-referenced audio," and "Head-referenced audio."

Figure 10-2 shows the applications in the book that use three-dimensional audio.

One more example is speech recognition. One can query based on the general characteristic of "speech recognition." Beyond that, one can also query about various characteristics of speech recognition. The subcategories include "Speaker independent," "Speaker dependent," "Push-to-talk," "Name-to-talk," and "Look-to-talk." It is important to note that the subcategories are not simply a list from which you can choose only one. For example, an application could use speech recognition—that is, "Speaker dependent"—that uses the "Name-to-talk" method of invoking it. If a subcategory is invoked without specifying one of the attributes, the results will be returned as though all elements in that subcategory were specified. So, if a query is made to return all applications that use speech recognition that is invoked by the "Name-to-talk" method, the results will include all applications that include "Name-to-talk" regardless of whether the speech recognition is "Speaker independent," or "Speaker dependent."

In the applications in this book, the two applications that use speech recognition are *Officer of the Deck*, and *Virtual Director*. Refining the query to seeking applications that use speaker-independent systems limits the results to the only application that fulfills the criteria, *Officer of the Deck*.

Figure 10-3 shows the relationship between the two applications that use speech recognition. The diagram shows that both applications use speech recognition, they both use the "push-to-talk" method to invoke speech recognition, but also illustrates that *Officer of the Deck* uses speaker independent recognition whereas *Virtual Director* uses speaker dependent speech recognition.

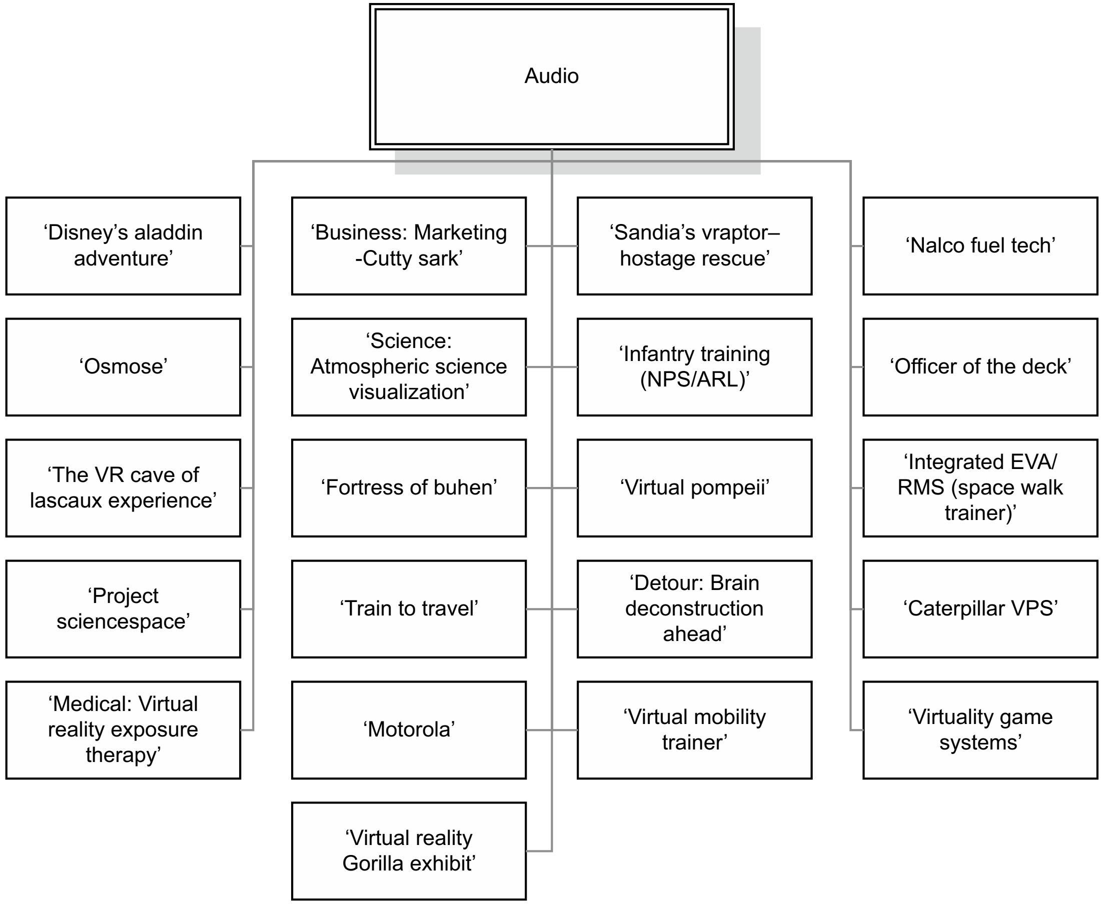

FIGURE 10-1
This diagram depicts the applications in this book that use audio.

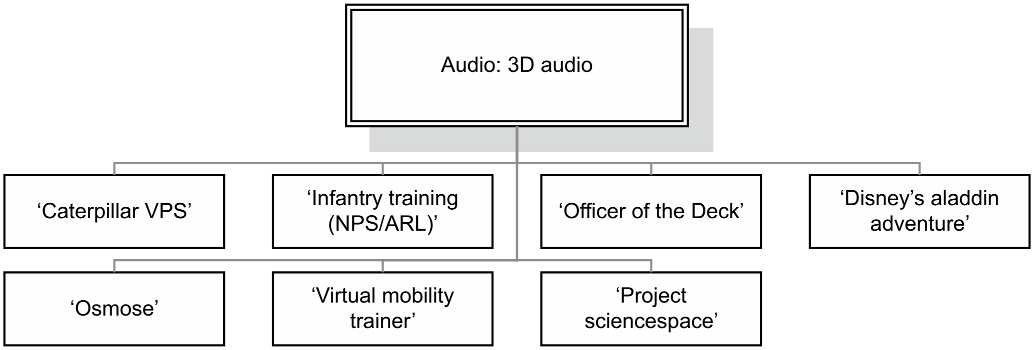

FIGURE 10-2

This diagram shows which applications in this book use three-dimensional audio.

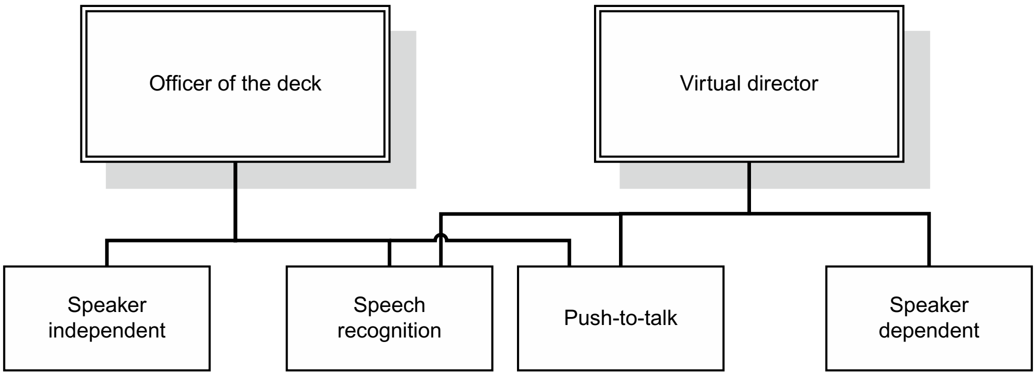

FIGURE 10-3

This diagram shows the relationship between the two applications in this book that use speech recognition.

10.4 FINDING APPLICATIONS WITH SPECIFIC CHARACTERISTICS

One of the most helpful ways to use the companion database is as an aid in identifying specific applications that possess the characteristics you are interested in. So, imagine you are considering creating an application that uses a head-mounted display, and you are planning to use it in a public venue with a large throughput of end users. By querying the database for applications with those traits, you would learn which applications fulfill those criteria. In this particular example, one of the applications that would be returned is Disney's *Aladdin Adventure*. Even though your goal was not to create an entertainment application that was for a theme park, you could benefit from the *Aladdin* application developers' experience in using head-mounted displays in a public setting. Their experience with the durability, hygiene, and "suit-up" time issues of head-mounted displays in that setting could help you avoid similar trial and error when you deploy your application.

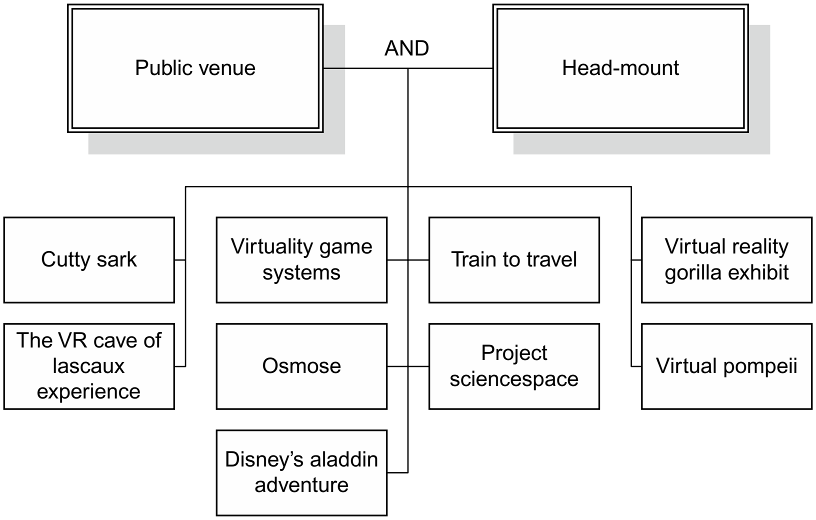

FIGURE 10-4
The diagram above illustrates which applications described in this book use head-mounted displays in public venues.

As an example of the scenario described above, the following diagram lists the applications described in this book that use head-mounted displays in venues that are public (theme parks, museums, etc.).

10.4.1 Showing all the characteristics of a specific application

It is often helpful to have a list of all the characteristics of a specific VR application. One reason such a list is beneficial is that it shows some of the characteristics that are commonly found together in successful applications. For example, by viewing a list of all the characteristics of the *Officer of the Deck* application, one can see that "push-to-talk" speech recognition is used in an application that uses a platform. By then studying the application write-up, it becomes evident that in that application end users hold the "push-to-talk" button in one hand while steadying themselves by holding the platform rail with the other.

By querying the online database for the traits of the *Officer of the Deck* application, you can see a list of the characteristics of that application. An example of the results of that query is shown below.

Another reason one might desire to create a list of all the characteristics of an application is to see what elements are commonly used in applications from a certain field. For example, one might want to see the characteristics of all the medical applications that are covered in this book. By querying the database for all "medical" applications, the following results are obtained.

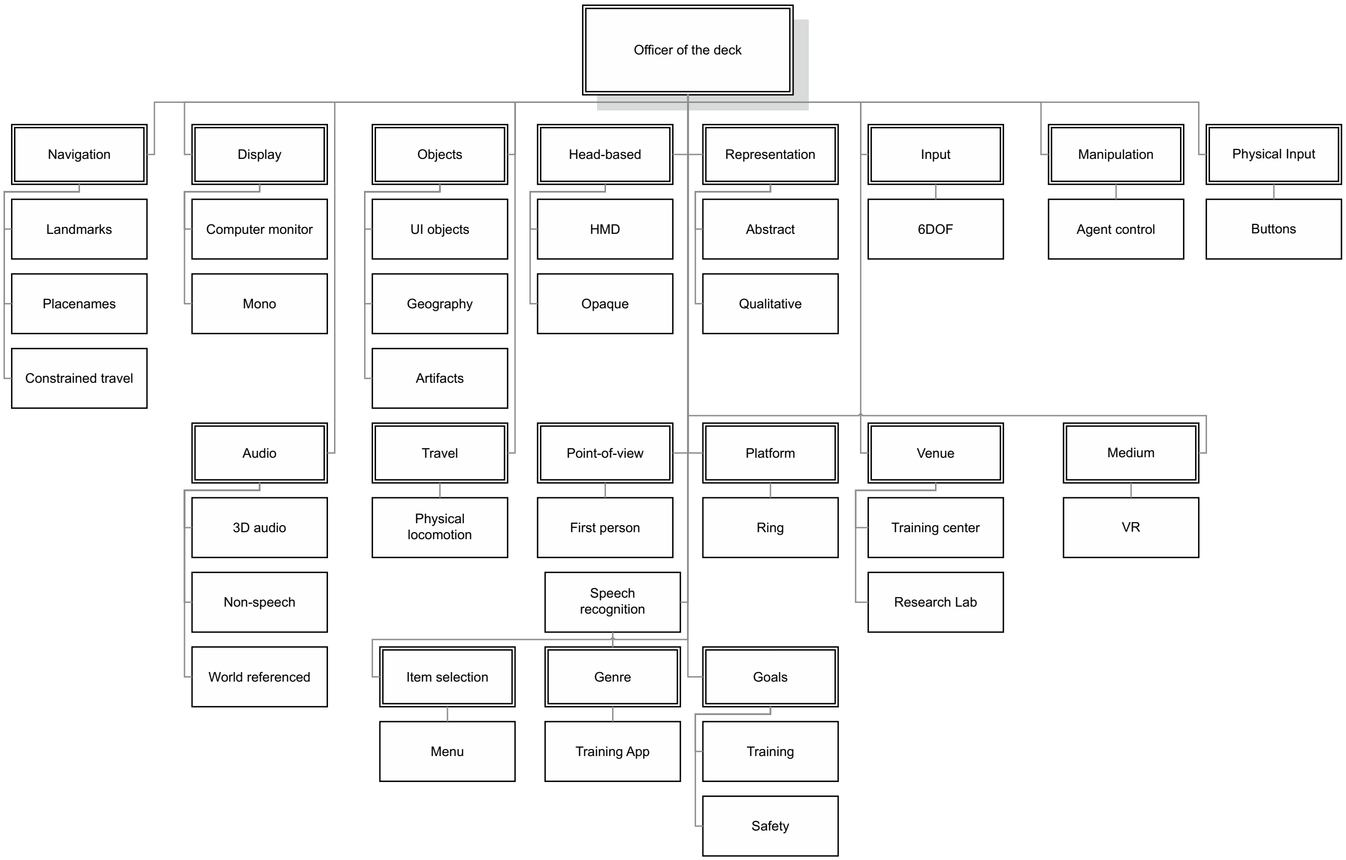

FIGURE 10-5

The diagram above shows the various characteristics and techniques that are used in the Officer of the Deck *application.*

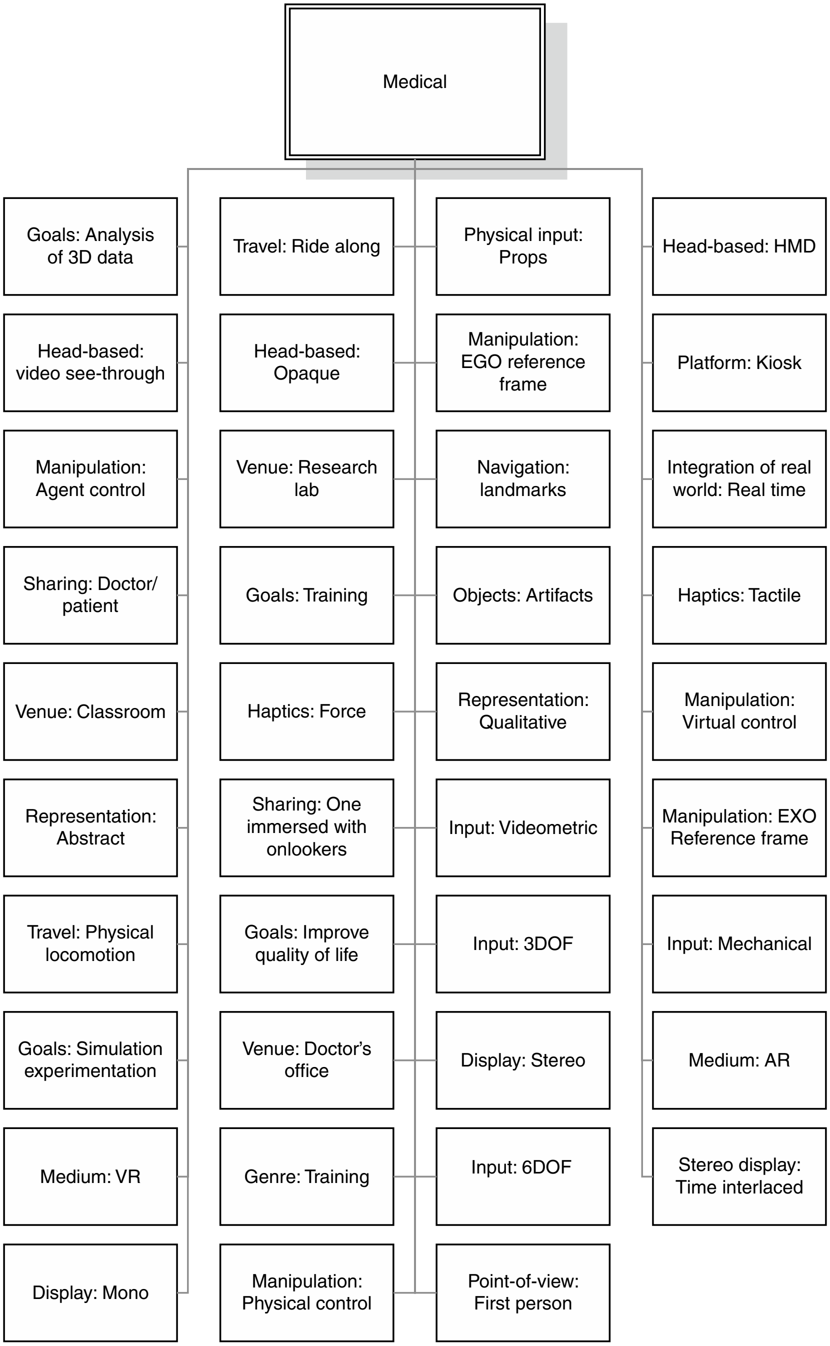

FIGURE 10-6

This image shows the results of a query to the database for a list of the characteristics of the union of all the medical applications described in this book.

10.4.2 Application taxonomies

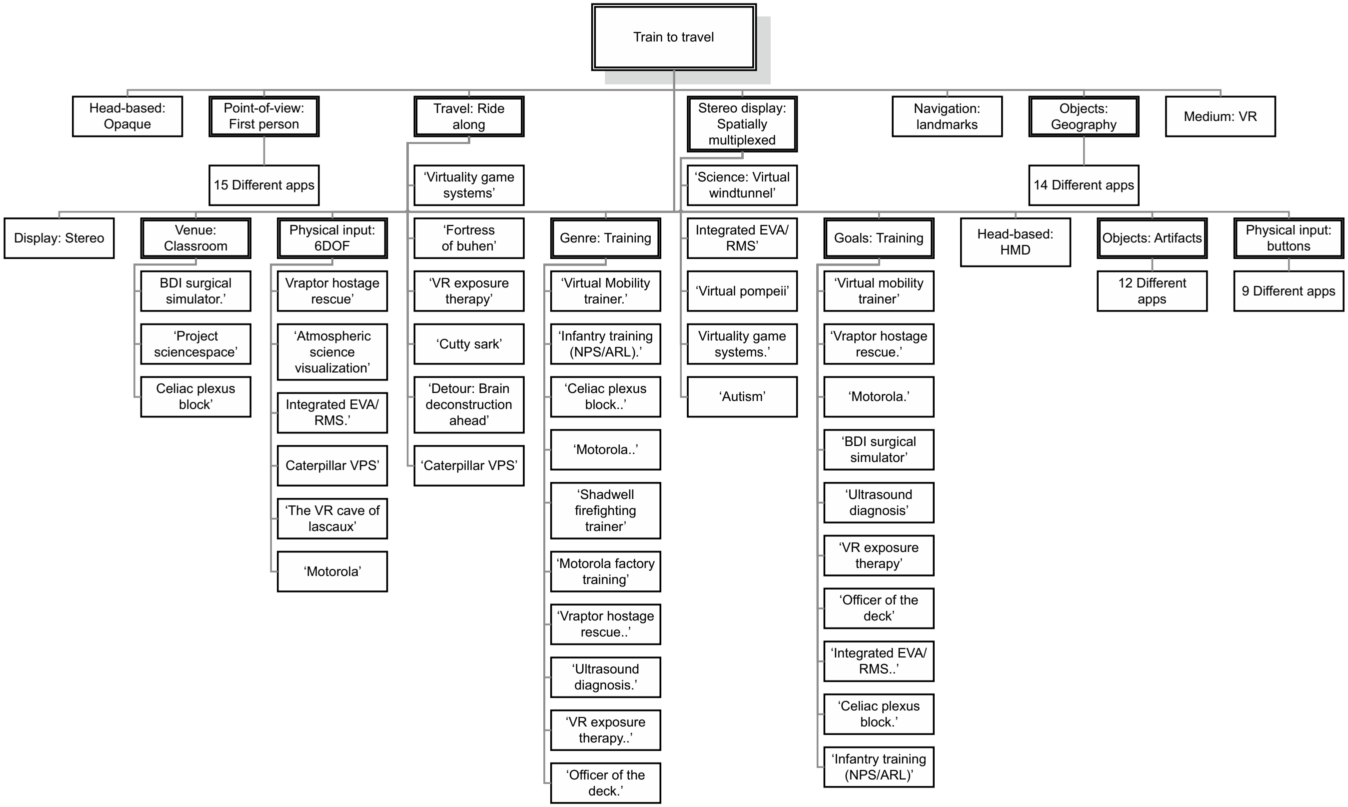

FIGURE 10-7

A second-level diagram, showing an application, its characteristics, and the related applications having each characteristic. For characteristics that have too many related apps to list, these have been omitted for space purposes.

The final example we show is a way to create taxonomies given the database access. Suppose you were especially interested in a particular application. You could list all of that application's characteristics. But then you may wish to know all the other applications that also have that attribute. In Figure 10-7, we show a diagram of "Train to Travel," and all of its characteristics. Then, under each of those characteristics, the other applications described in this book that share that characteristic are listed. In this way, the database can guide developers in shared "learned lessons" between seemingly unrelated applications.